信号与系统

School of Electronic
and Information Engineering

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2026.3

1. Introduction to Signal and System Analysis

- Introduction
- Signal description and classification
- System description and classification
- Applications of signals and systems

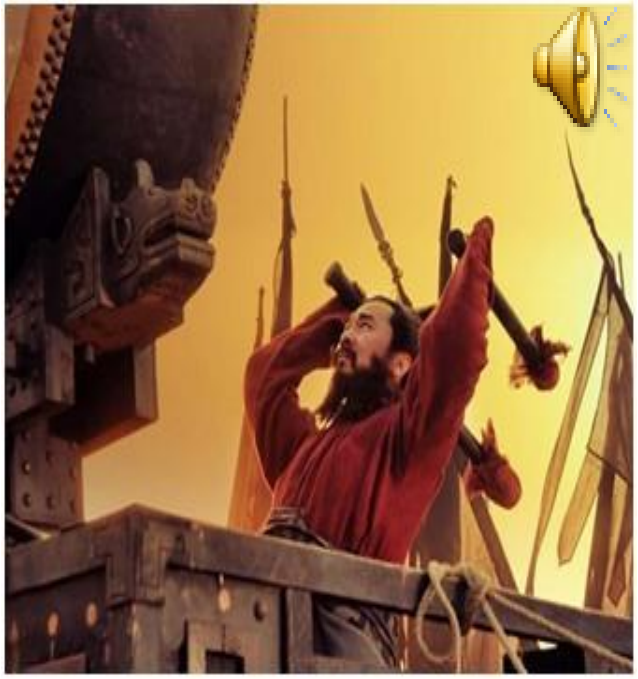
Description of the signal

- ※ delivery of messages
- ※ signal definition
- ※ signal representation

1. delivery of messages

- How is the message delivered?

- using the form of sound



1. delivery of messages

- How is the message delivered?

  - using the form of light

Beacon

Signal flares

Lighthouse

Traffic signals

Railroad Signal Light

1. delivery of messages

- How is the message delivered?

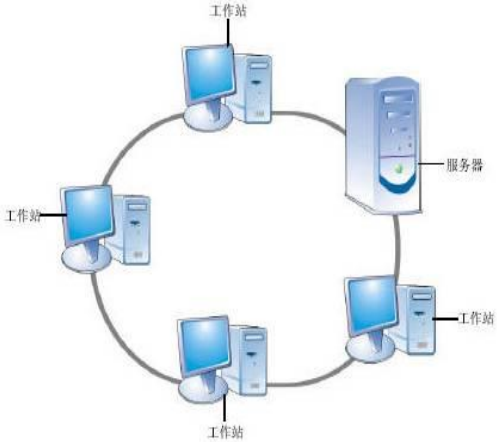
- using the form of electricity

Transmitter

Telephone

Pager

Mobile Phone



Computer Networks

Virtual Wi-Fi pulse

2. signal definition

Message → Forms of sound, light, electricity, … → Message

People usually use sound, light, and electricity as carriers to transmit or receive messages.

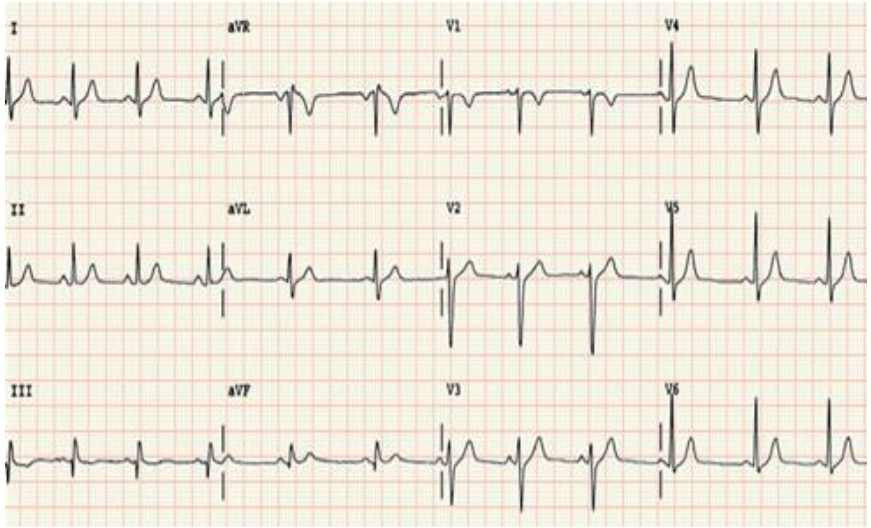
- Definition 1: A signal is the manifestation or transmission carrier of a message (narrow sense)

  A message is the specific content of a signal.

2. signal definition

- Definition 1: A signal is the manifestation or transmission carrier of a message (narrow sense)

Electrocardiogram



1. Diagnosis of cardiac arrhythmias and conduction disturbances
2. Diagnosis of myocardial infarction and the range of diseased stages
3. Diagnosis of cardiomyopathy, coronary insufficiency...

Consumer Price Index

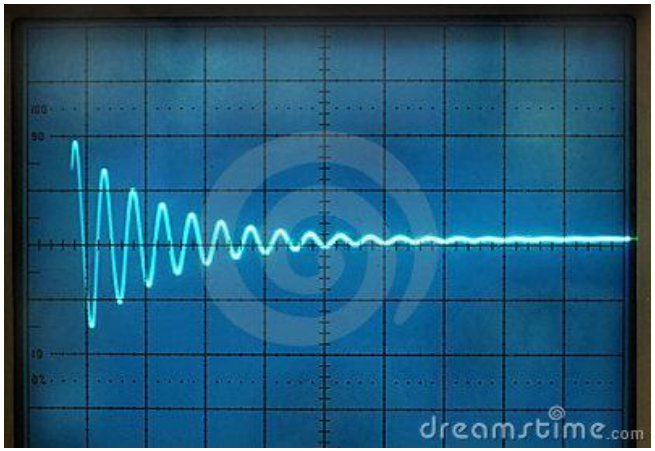
1. Reflect the level of inflation
2. Reflect changes in the purchasing power of currency
3. Reflect the impact on real wages of employees

2. signal definition

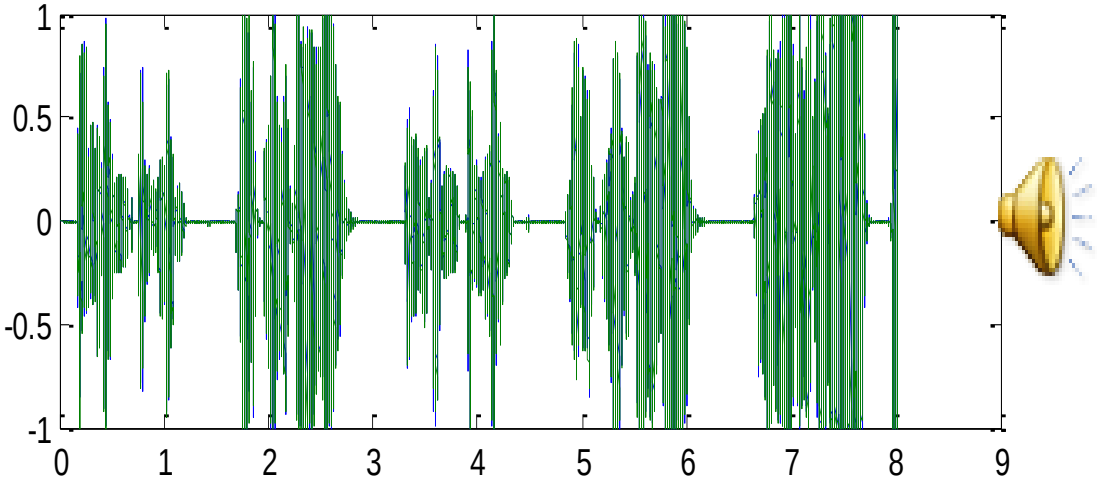
- Definition 2: A signal is a physical quantity that varies with some parameters (in a broad sense)

Electrical signals are usually represented as voltage or current (charge or flux) over time

Speech signal as a function of air pressure over time



Current observed with an oscilloscope



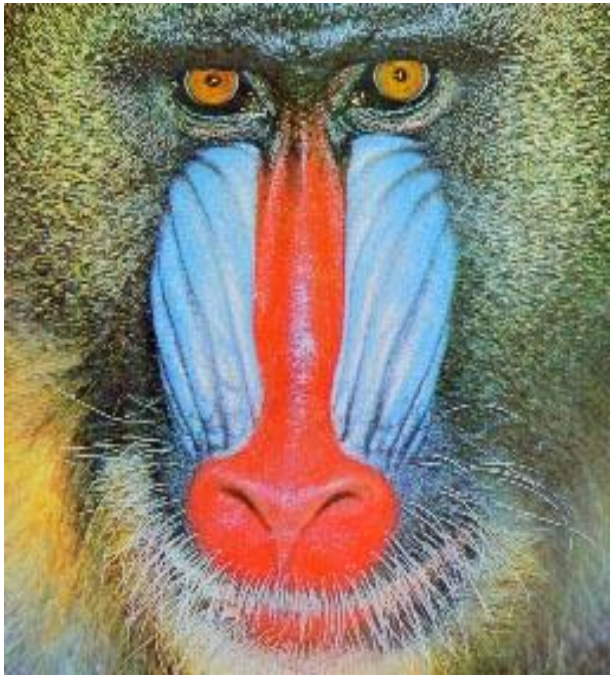
Child's voice answering the phone

2. signal definition

black and white image signal

A signal whose brightness varies with spatial position (x,y)

color image signal



$$I(x,y)=\begin{bmatrix} I_R(x,y) \\ I_G(x,y) \\ I_B(x,y) \end{bmatrix}$$

Three primary colors red (R), green (G), blue (B)
A signal that varies with spatial position (x,y)

3. signal representation

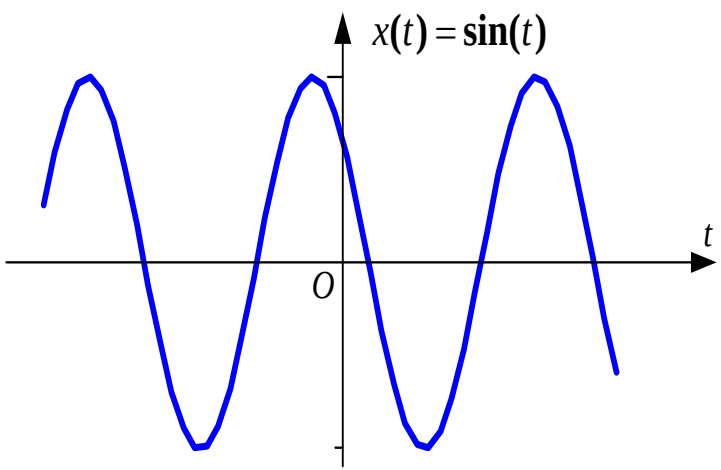
(1) Mathematical Expressions - Functions

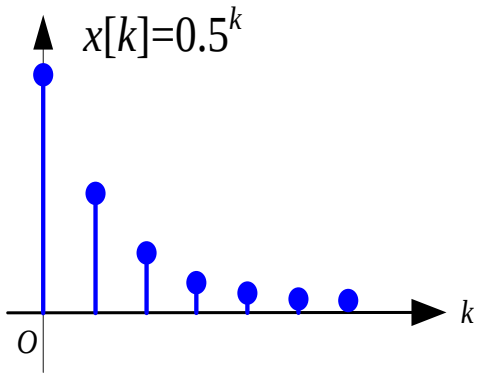
General： $x(t)$、$x[k]$

Specific： $x(t)=\sin(t)$、 $x[k]=0.5^k$

In this lesson, "signal" and "function" are equivalent.

(2) Waveform



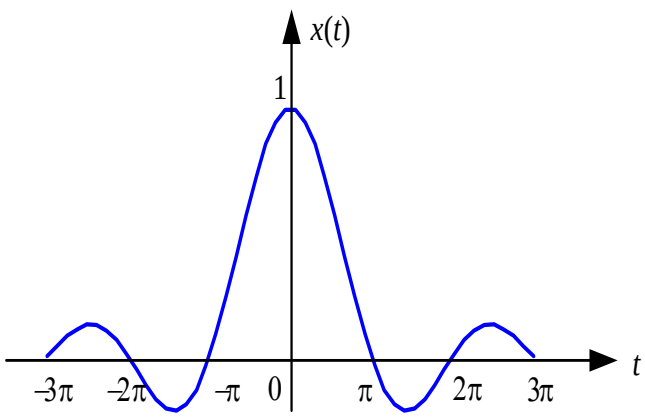


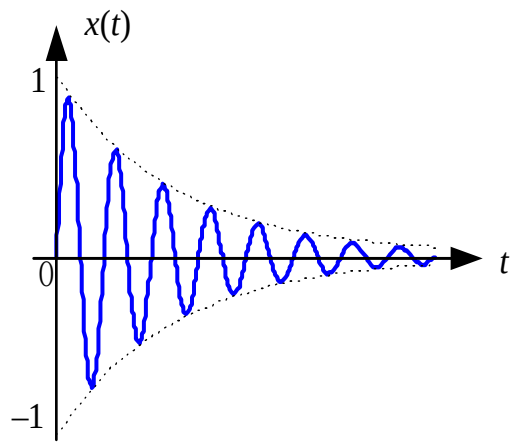
Classification of signals

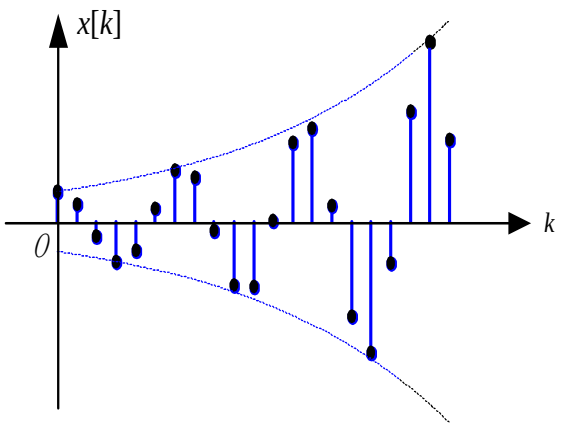
- Deterministic Signals vs Random Signals
- continuous-time signal vs. discrete-time signal
- periodic signal and non-periodic signal
- energy signal and power signal

1. Deterministic vs Random Signals

- Deterministic signal: A signal has a definite value at every point on the domain.







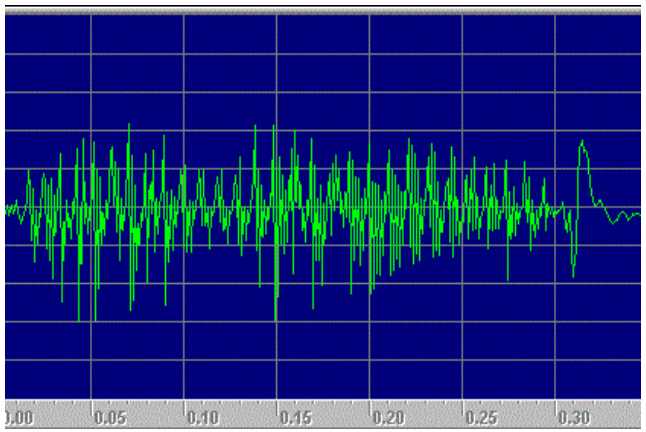
$$x(t)=\frac{\sin t}{t}, t\in \mathrm{R}$$

$$x(t)=\mathrm{e}^{-t}\sin t, t>0$$

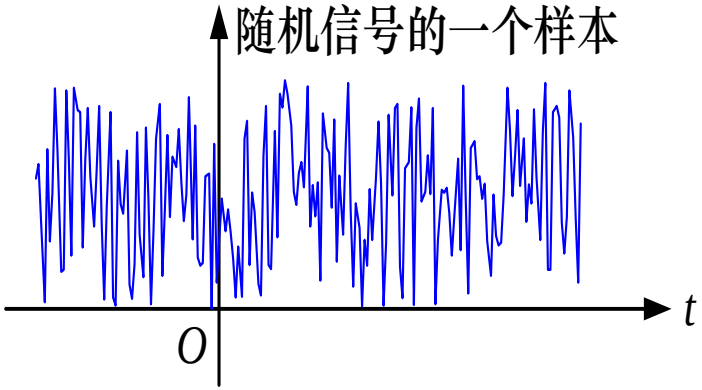
$$x[k]=2^{k}\sin k, k\in \mathrm{I}$$

1. Deterministic vs Random Signals

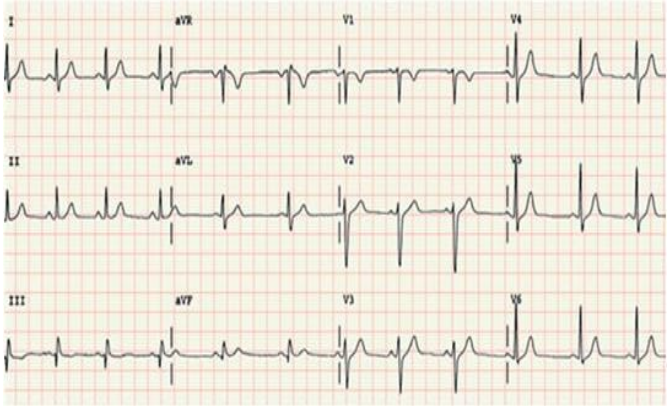
- Random signal: A signal that does not meet the definition of a deterministic signal.



voice signal



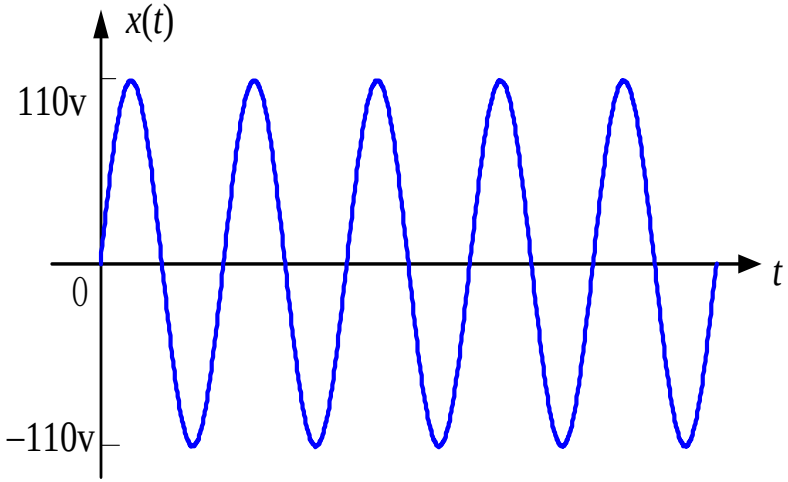
noise signal



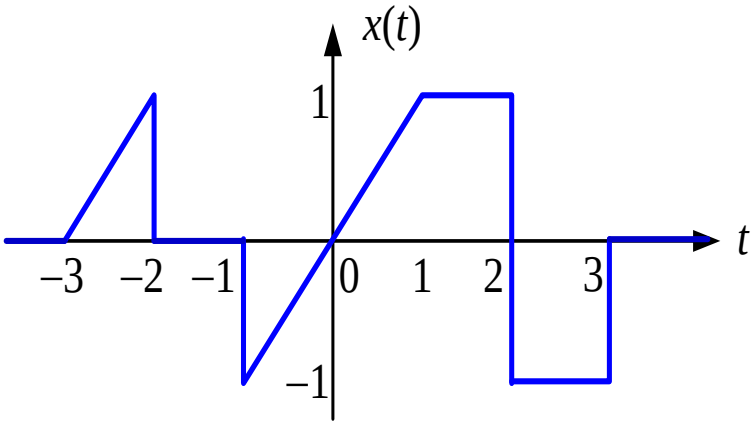
ECG signal

2. Continuous Time Signals vs. Discrete Time Signals

- **Continuous-time signal**: the domain of the signal is a continuous interval, usually represented by x(t), t∈R.



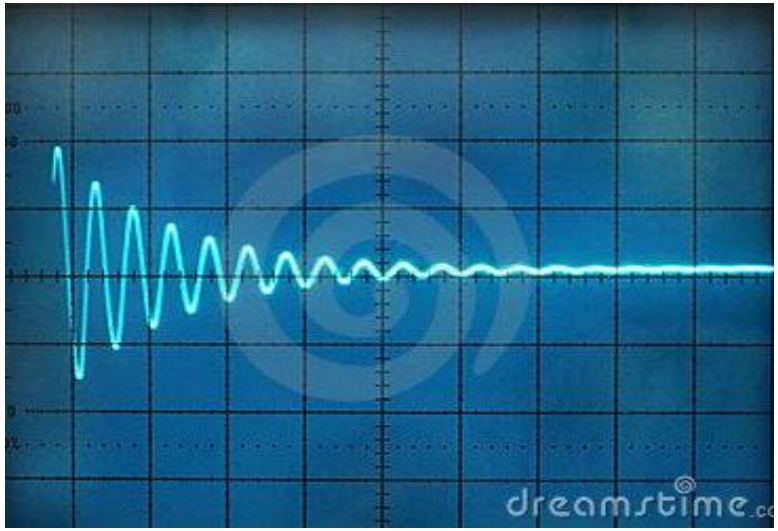
The domain of definition is a continuous interval



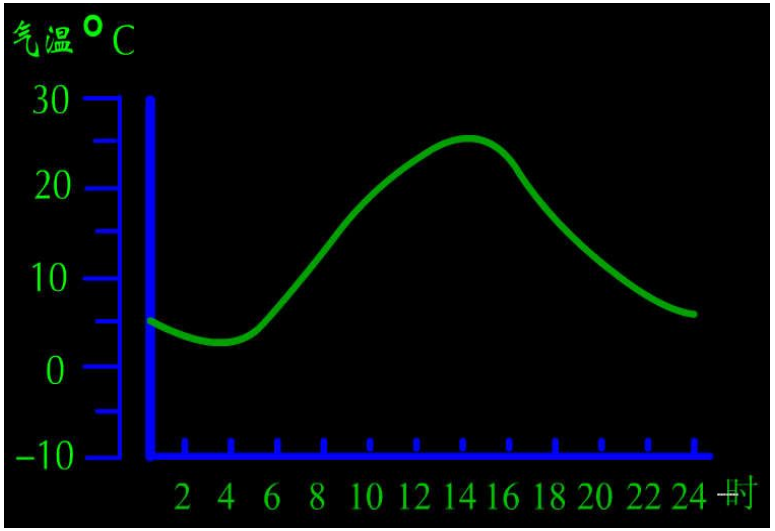
allowed on its domain
There are a finite number of discontinuities.

2. Continuous Time Signals vs. Discrete Time Signals

- Continuous-time signal: The domain of the signal is a continuous interval.



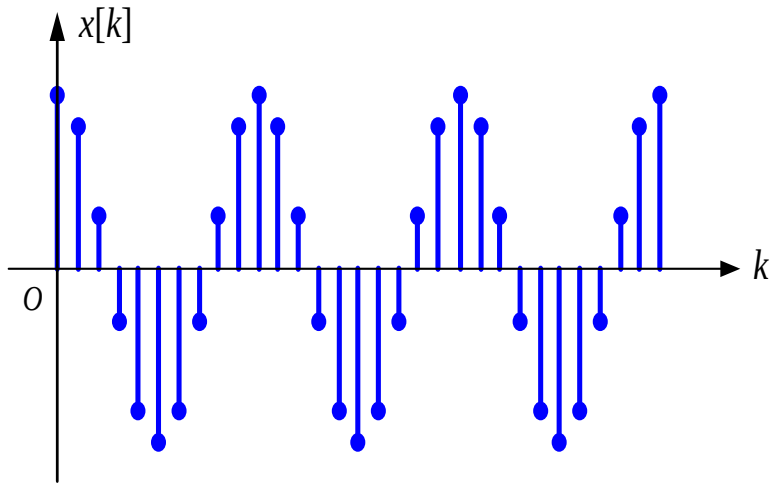
current signal

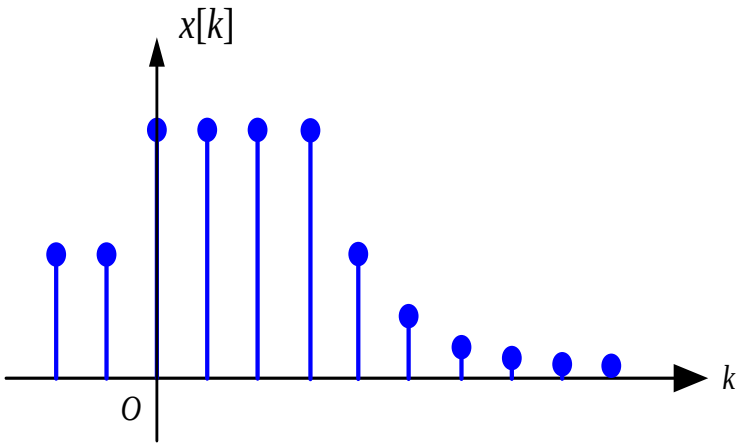


temperature signal

2. Continuous Time Signals vs. Discrete Time Signals

- **Discrete-time signal**: The domain of the signal is some discrete points, usually represented by x[k], k ∈ I.





The domain of definition is a discrete point

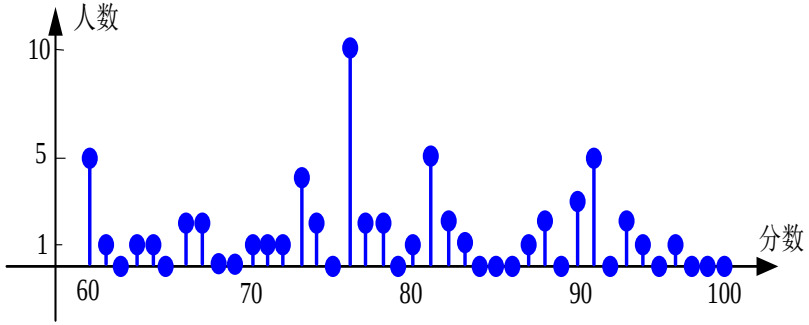
(When drawing a waveform diagram, the vertical axis can also be omitted)

2. Continuous Time Signals vs. Discrete Time Signals

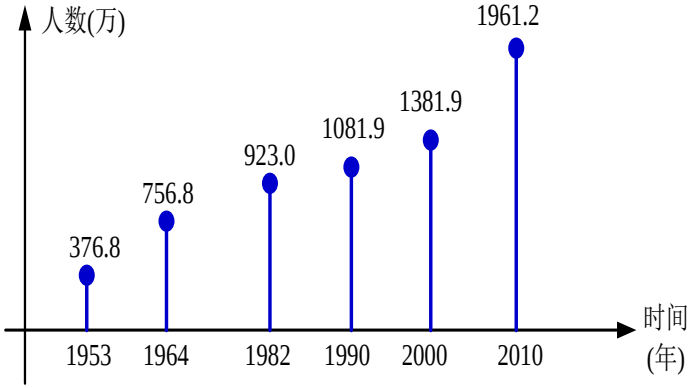
- **Discrete-time signal**: The domain of the signal is some discrete points.

Consumer Price Index

Composite Index



Grade distribution

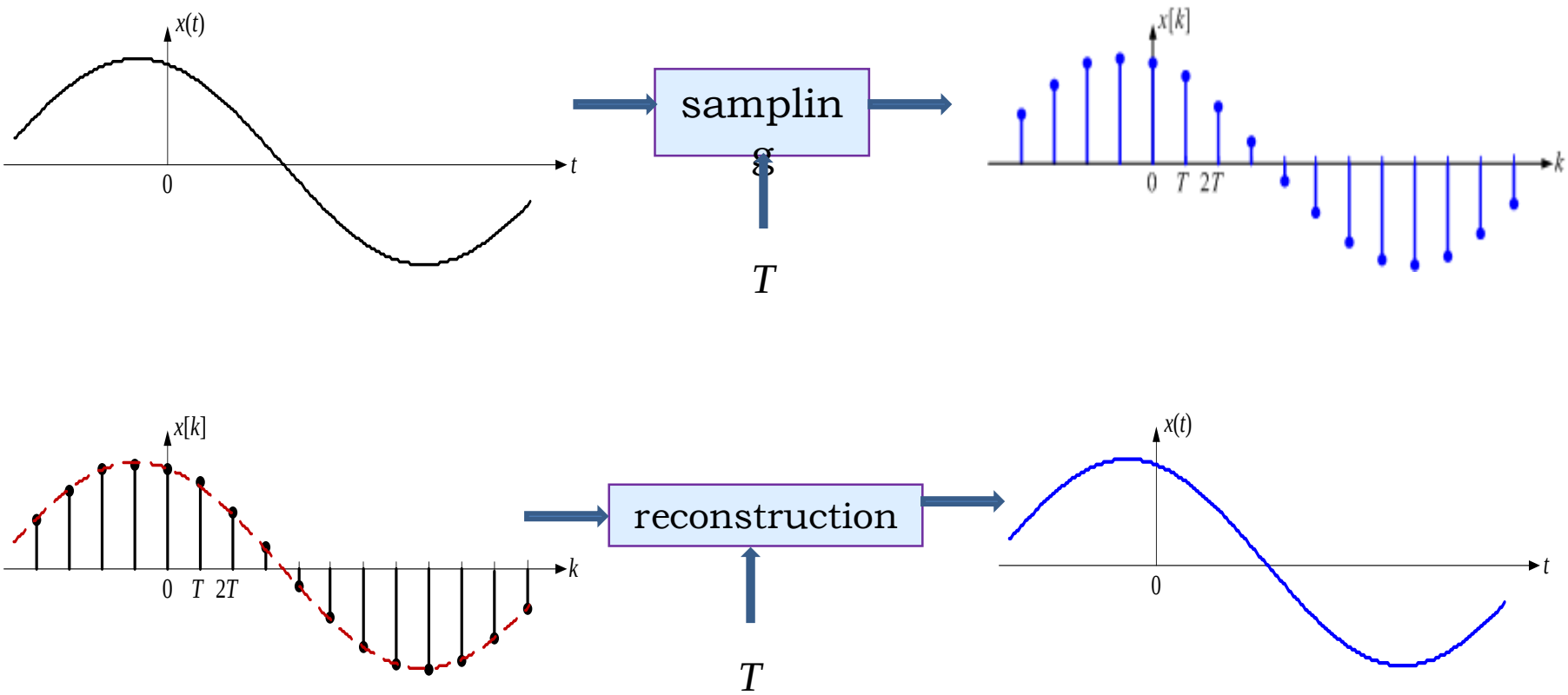


Permanent Population of Beijing

2. Continuous Time Signals vs. Discrete Time Signals

- relationship between the two

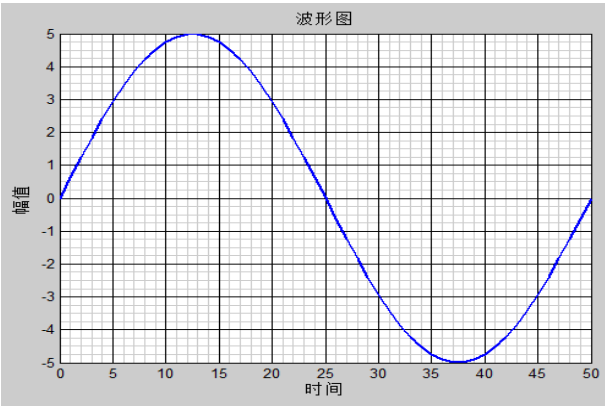
$$x(t)\big|_{t=kT} = x[k]$$



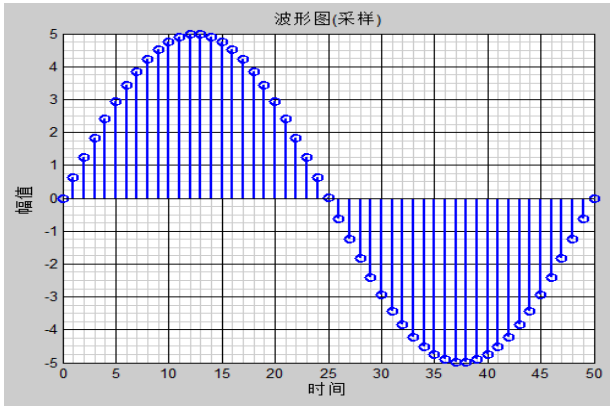
2. Continuous Time Signals vs. Discrete Time Signals

Analog and Digital Signals

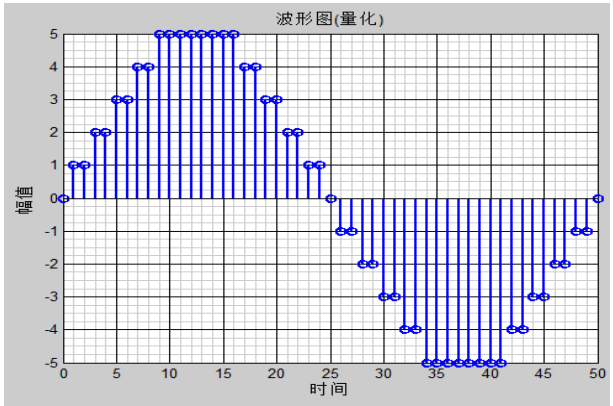
- Analog signal: A continuous-time signal with continuous amplitudes.
- Digital Signal: A discrete-time signal with discrete amplitudes.



analog signal



discrete signal



Digital signal

3. Periodic and non-periodic signals

- continuous time periodic signal: $\forall t \in \mathrm{R}$, There are positive numbers $\boldsymbol{T}$, makes

$$x(t+T)=x(t)$$

set up,Then x(t) is a periodic signal.

- discrete-time periodic signal: $\forall k \in \mathrm{I}$, There are positive integers $\boldsymbol{N}$, makes

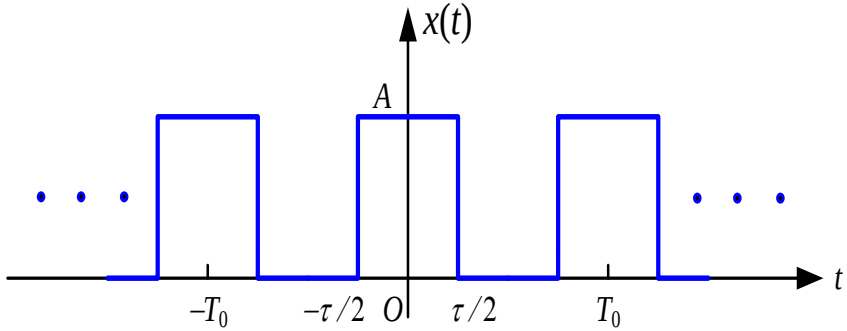
$$x[k+N]=x[k]$$

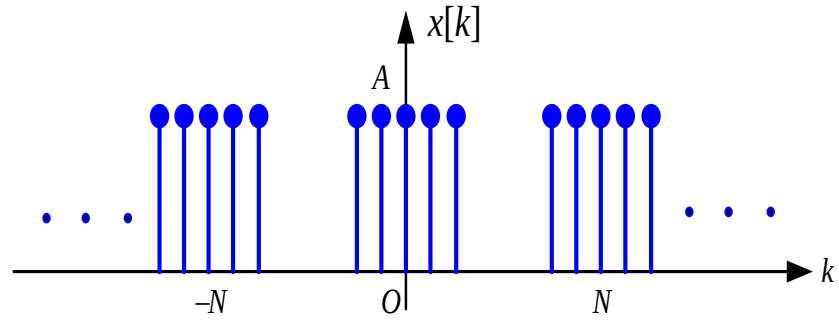
set up,Then $x[k]$ is a periodic signal.

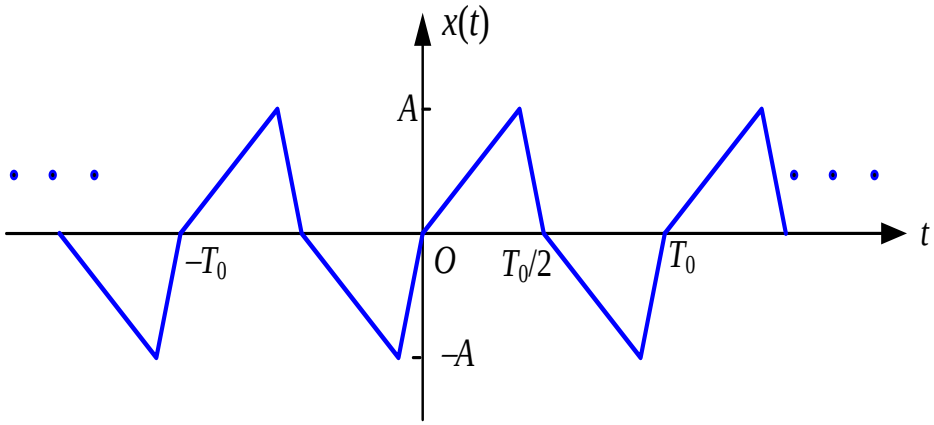
The smallest positive T and positive N that meet the above conditions are called the fundamental period of the signal.

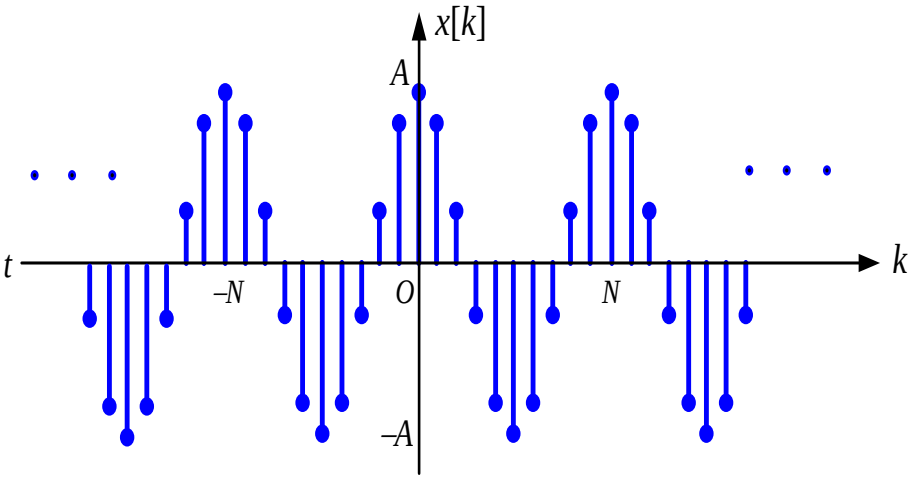
- Non-periodic signal: A signal that does not satisfy the definition of a periodic signal.

3. Periodic and non-periodic signals





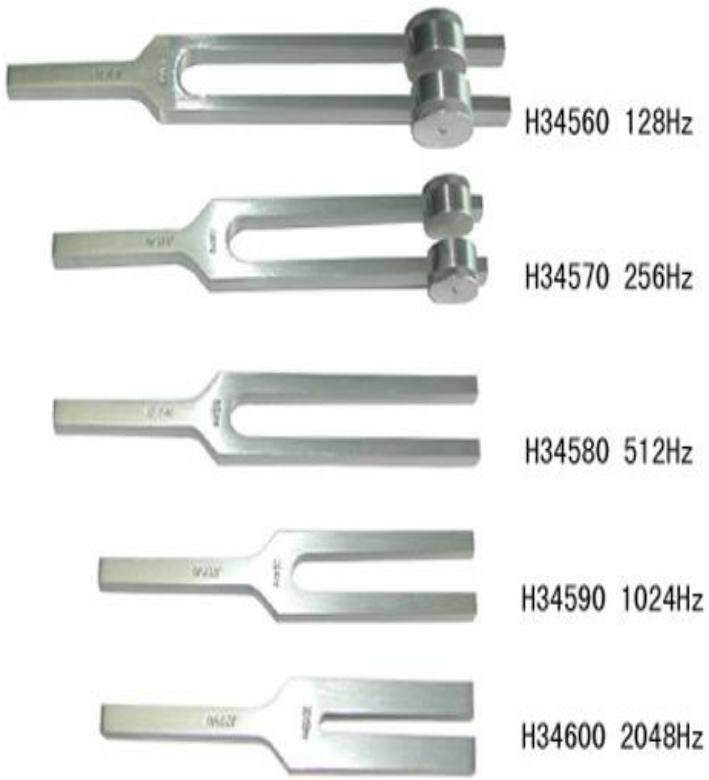




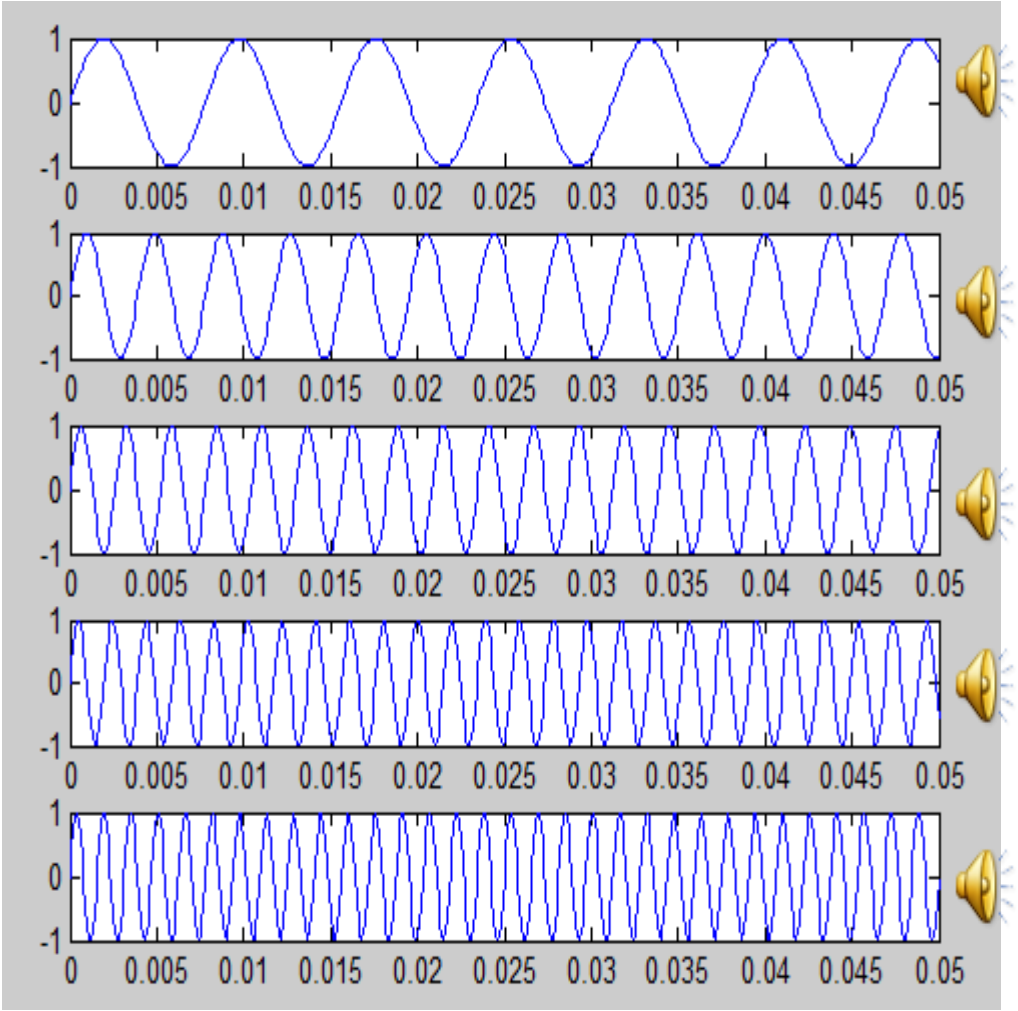
continuous time period signal

discrete-time periodic signal

3. Periodic and non-periodic signals



audiometric tuning fork



The corresponding waveform of the signal sent by the tuning fork

3. Periodic and non-periodic signals

Example 1: Identify sinusoidal signal $x(t)=\sin(\omega_0 t)$ Whether it is a periodic signal.

Solution:

Beceuse $x(t+T)=\sin\omega_0(t+T) \ =\sin(\omega_0 t+\omega_0 T)$

To make it a periodic signal, there must be

$$\omega_0 T = m2\pi \ , \quad m\text{=integer}$$

$$\Downarrow$$

$$T = m\frac{2\pi}{\omega_0} \ , \quad m\text{=integer}$$

Continuous sine signals are periodic signals!

Thus, $\sin(\omega_0 t)$ is a periodic signal with period $2\pi/|\omega_0|$.

3. Periodic and non-periodic signals

Example 2: Identify a discrete sine signal $x[k]=\sin(\Omega_0 k)$ whether it is a periodic signal.

Solution:

Beceuse $x[k+N]=\sin\Omega_0(k+N)=\sin(\Omega_0 k+\Omega_0 N)$

To make it a periodic signal, there must be

$$\Omega_0 N = m2\pi \ , \quad m\text{=integer}$$

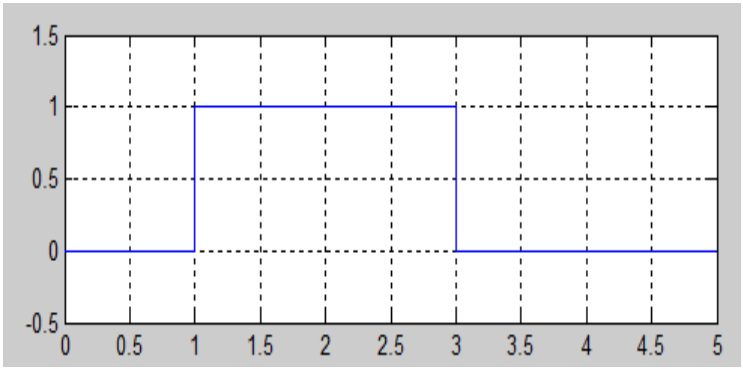
A discrete sine signal is not necessarily a periodic signal!

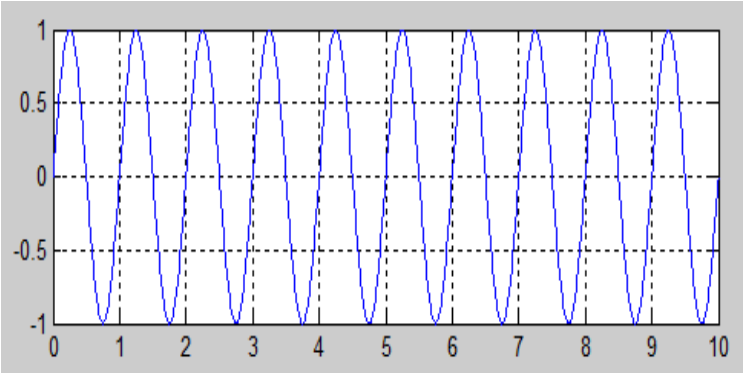
$$\frac{|\Omega_0|}{2\pi}=\frac{m}{N}$$ =Rational numbers, m and N are both integers

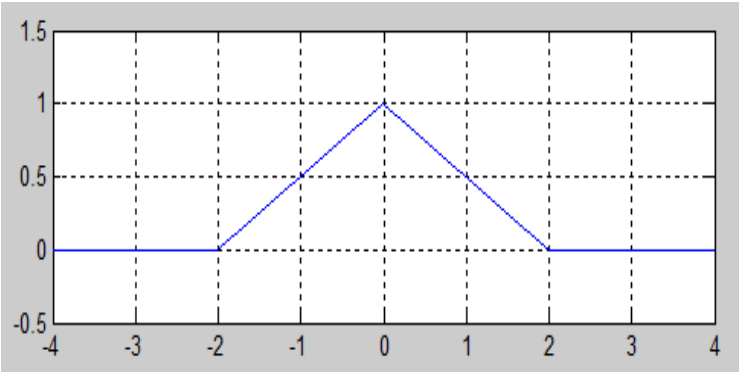
Thus, $|\Omega_0|/2\pi$ is a periodic signal only when it could be expressed as a rational number.

4. Energy Signal vs. Power Signal

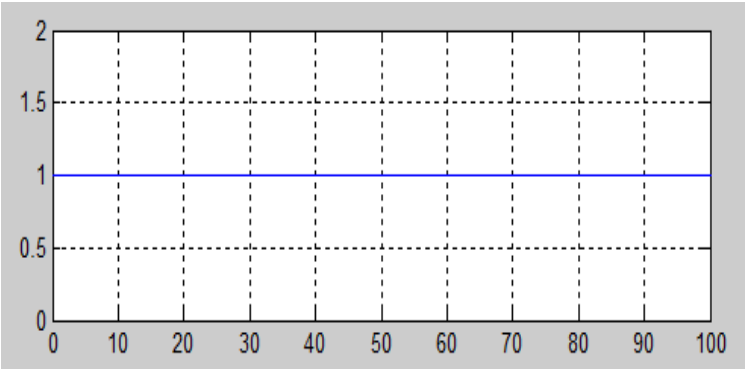
- Energy signal: Normalized energy is a non-zero finite value, normalized power is zero.
- Power signal: normalized energy to an infinite value, normalized power to a nonzero finite value.







energy signal



power signal

4. Energy Signal vs. Power Signal

- Energy signal: Normalized energy is a non-zero finite value, normalized power is zero.
- Power signal: normalized energy to an infinite value, normalized power to a nonzero finite value.

Definition of normalized energy W and normalized power P:

Continuous signal

$$W = \lim_{T\to\infty}\int_{-T/2}^{T/2}\left|x(t)\right|^2 \mathrm{d}t$$

$$P = \lim_{T\to\infty}\frac{1}{T}\int_{-T/2}^{T/2}\left|x(t)\right|^2 \mathrm{d}t$$

Discrete signal

$$W = \lim_{N\to\infty}\sum_{-N}^{N}\left|x[k]\right|^2$$

$$P = \lim_{N\to\infty}\frac{1}{2N+1}\sum_{-N}^{N}\left|x[k]\right|^2$$

4. Energy Signal vs. Power Signal

Example 3: Identify a periodic signal $x(t)=\cos(\omega_0 t)$

Is it energy signal or power signal?

Solution:

$\cos(\omega_0 t)$is the periodic signal with fundamental period $T_0=2\pi/|\omega_0|$.

Its energy in one fundamental period is

$$E_0=\int_0^{T_0}|x(t)|^2\mathrm{d}t=\int_0^{T_0}\cos^2(\omega_0 t)\mathrm{d}t=\frac{T_0}{2}$$

Since periodic signals have infinite periods, the normalized energy E has infinite values.

The normalized power is

$$P=\lim_{T\to\infty}\frac{1}{T}\int_{-T/2}^{T/2}|x(t)|^2\mathrm{d}t=\lim_{n\to\infty}\frac{1}{nT_0}nE_0=\frac{1}{2}$$

The normalized power P is a non-zero finite value and therefore a power signal.

4. Energy Signal vs. Power Signal

Example 4: Determine whether the DC signal $x(t)=C$ (C is a constant) is an energy signal or a power signal.

Solution: The normalized energy is

$$E=\lim_{T\to\infty}\int_{-T/2}^{T/2}|x(t)|^2\mathrm{d}t=\lim_{T\to\infty}\int_{-T/2}^{T/2}C^2\mathrm{d}t=\lim_{T\to\infty}C^2T=\infty$$

The normalized power is

$$P=\lim_{T\to\infty}\frac{1}{T}\int_{-T/2}^{T/2}|x(t)|^2\mathrm{d}t=\lim_{T\to\infty}\frac{1}{T}\int_{-T/2}^{T/2}C^2\mathrm{d}t=\lim_{T\to\infty}\frac{C^2T}{T}=C^2$$

E is an infinite value, while P is a finite value and therefore a power signal.

Both periodic and DC signals are power signals.

4. Energy Signal vs. Power Signal

Example 5: Determine whether $x[k]=0.5^k$ is the energy signal or power signal.

Solution: The normalized energy is

$$E=\lim_{N\to\infty}\sum_{k=-N}^{N}|x[k]|^2=\lim_{N\to\infty}\sum_{k=-N}^{N}(0.5)^{2k}=\infty$$

The normalized power is

$$P=\lim_{N\to\infty}\frac{1}{2N+1}\sum_{k=-N}^{N}\left(\frac{1}{2}\right)^{2k}=\infty$$

E and P are both infinite value, therefore it is neither an energy signal nor a power signal.

Note: A signal can be neither an energy signal nor a power signal, but it cannot be both an energy signal and a power signal.

Thanks

Some of the materials quoted in this course are accumulated by the main lecturer over the years, from a variety of media and colleagues, peers and friends of the exchange, it is difficult to attribute them all, so I would like to state and thank you.